

DNA is the genetic material that stores, replicates and transmits genetic information.

MOLECULAR BASIS OF INHERITANCE

OVERVIEW

This chapter explains how genetic information is stored in DNA, how it is replicated and expressed to form proteins which control structure and function of an organism.

TOPICS COVERED

- DNA – The Genetic Material
- Structure of DNA
- DNA Replication
- Transcription
- Genetic Code
- Translation

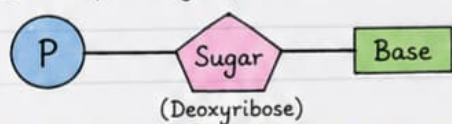
1. DNA – THE GENETIC MATERIAL

- In 1944, **Avery, MacLeod and McCarty** proved that DNA is the genetic material.
- **Hershey and Chase (1952)** confirmed using bacteriophage experiment.
- DNA is found in nucleus, mitochondria and plastids.
- It is a long polymer of **nucleotides**.

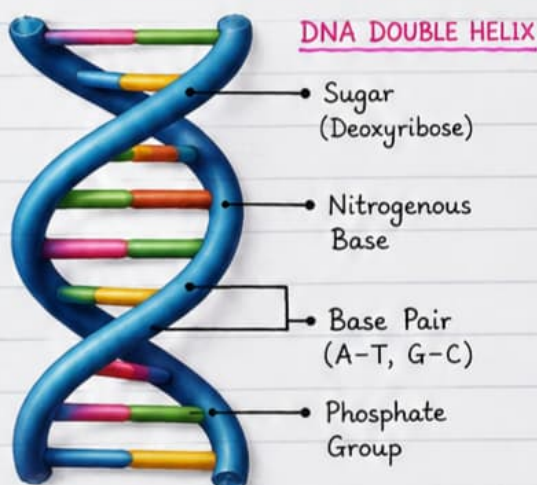
NUCLEOTIDE – THE BUILDING UNIT

Each nucleotide has three components:

- ① **Nitrogenous base** – Purine (A, G)
Pyrimidine (T, C)
- ② **Pentose sugar** – Deoxyribose
- ③ **Phosphate group** – PO_4^{3-}



"Nucleotide = Base + Sugar + Phosphate"



COMPLEMENTARY BASE PAIRING

Adenine (A) pairs with Thymine (T)

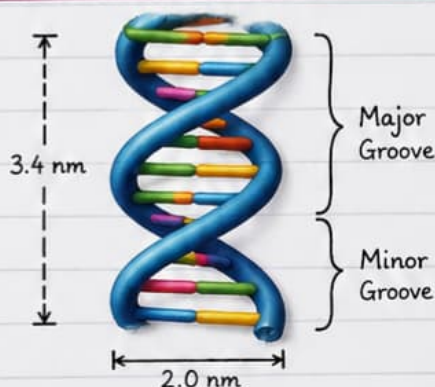


Guanine (G) pairs with Cytosine (C)



2. STRUCTURE OF DNA

- Discovered by **Watson and Crick (1953)**
- DNA is a double helix with two **antiparallel** strands.
- Backbone is made of **Sugar-Phosphate**.
- Bases are inside and backbone is outside.
- It is right handed helix.
- Distance between two consecutive base pairs = **0.34 nm**
- 10 base pairs make one complete turn (**3.4 nm**).
- **Major groove** and **Minor groove** are present.



★ IMPORTANT POINTS

- DNA is the genetic material.
- It is made of nucleotides.
- $A = T$ and $G \equiv C$
- DNA replicates semiconservatively.
- Information flows: DNA → RNA → Protein



"From DNA to life,
the journey of
information never ends."

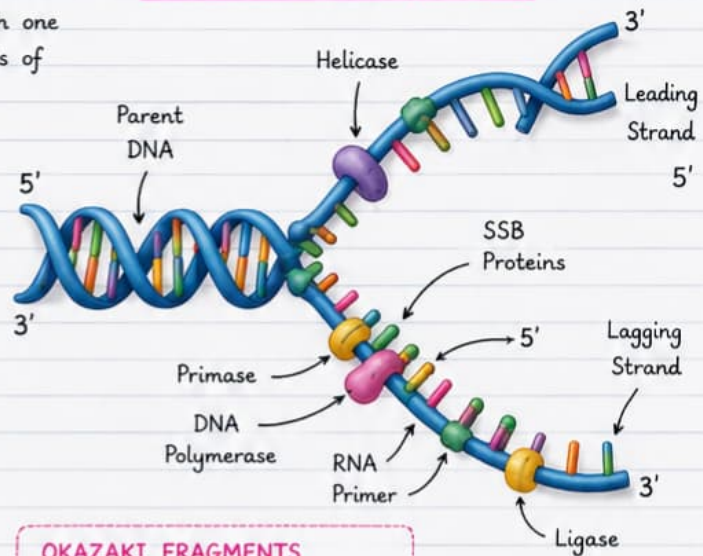
3. DNA REPLICATION

- ☆ Replication is a **semi-conservative** process in which one parental strand serves as a template for synthesis of a new complementary strand.

KEY FEATURES

- Semi-conservative
- Bidirectional (in both directions)
- Semi-discontinuous (leading & lagging strands)
- Requires primer
- Occurs in 5' → 3' direction
- Enzymes required: DNA polymerase, Helicase, Primase, Ligase, SSB proteins

REPLICATION FORK (3D VIEW)



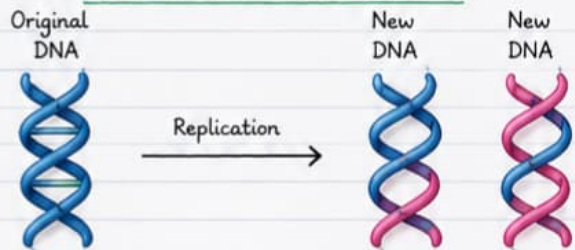
STEPS OF DNA REPLICATION

- 1 **Initiation:** Helicase unwinds the double helix at origin of replication.
- 2 **Unwinding:** SSB proteins stabilize the single strands.
- 3 **Primer synthesis:** Primase makes short RNA primers.
- 4 **Elongation:**
 - **Leading strand** → continuous synthesis towards 3' end.
 - **Lagging strand** → discontinuous synthesis away from 3' end (Okazaki fragments).
- 5 **Termination:** Ligase joins Okazaki fragments to form a continuous strand.

OKAZAKI FRAGMENTS

Short DNA fragments synthesized on the lagging strand. ☆

SEMI-CONSERVATIVE NATURE



Each daughter DNA has one old (parental) strand and one newly synthesized strand. ♥

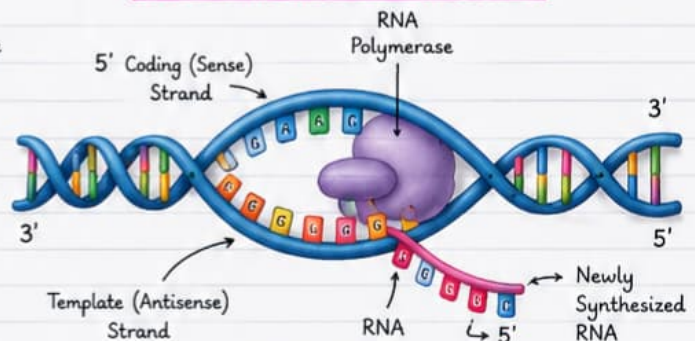
4. TRANSCRIPTION

- ☆ The process of synthesis of **RNA** from a DNA template is called transcription.

KEY POINTS

- Template strand of DNA is read in 3' → 5' direction.
- RNA is synthesized in 5' → 3' direction.
- Enzyme required: **RNA polymerase**
- Occurs in nucleus in eukaryotes.
- In prokaryotes, it occurs in cytoplasm.

TRANSCRIPTION BUBBLE (3D VIEW)



TYPES OF RNA

Type	Structure	Function	Example/Notes
mRNA		Carries genetic information from DNA to ribosome.	Messenger RNA
tRNA		Brings amino acids to ribosome.	Transfer RNA
rRNA		Forms structural part of ribosome.	Ribosomal RNA

REPLICATION vs TRANSCRIPTION

REPLICATION	TRANSCRIPTION
Occurs in nucleus	Occurs in nucleus
DNA → DNA	DNA → RNA
Entire DNA is copied	Only a gene is transcribed
Needs primer	No primer required
Enzyme: DNA polymerase	Enzyme: RNA polymerase



Remember: **Replication** makes a copy of DNA, while **Transcription** reads the code to make RNA. ♥

Molecular Basis of Inheritance

"DNA carries the information, RNA reads it, Protein executes it."

5. GENETIC CODE

- The genetic code is the sequence of bases in **mRNA** which codes for a particular amino acid in a protein.
- It was deciphered by **Nirenberg and Khorana (1960s)**.
- It is a **triplet code** (3 bases = 1 codon).
- There are **64 codons** (4^3) in total.
- **61 sense codons** code for 20 amino acids.
- **3 stop codons**: UAA, UAG, UGA (do not code for any amino acid).

CHARACTERISTICS OF GENETIC CODE

- Triplet
- Specific
- Degenerate
- Non-overlapping
- Commaless
- Universal
- Unambiguous
- Starts with AUG (start codon)

THE GENETIC CODE (mRNA CODON TABLE)

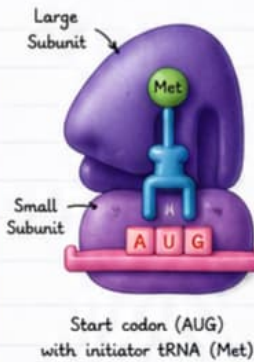
1st Base	2nd Base								3rd Base
	U		C		A		G		
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC	Phe	UCC	Ser	UAC	Tyr	UGC	Cys	C
	UUA	Leu	UCA	Ser	UAA	Stop	UGA	Stop	A
	UUG	Leu	UCG	Ser	UAG	Stop	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC	Leu	CCC	Pro	CAC	His	CGC	Arg	C
	CUA	Leu	CCA	Pro	CAA	Gln	CGA	Arg	A
	CUG	Leu	CCG	Pro	CAG	Gln	CGG	Arg	G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC	Ile	ACC	Thr	AAC	Asn	AGC	Ser	C
	AUA	Ile	ACA	Thr	AAA	Lys	AGA	Arg	A
	AUG	Met (Start)	ACG	Thr	AAG	Lys	AGG	Arg	G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC	Val	GCC	Ala	GAC	Asp	GGC	Gly	C
	GUA	Val	GCA	Ala	GAA	Glu	GGA	Gly	A
	GUG	Val	GCG	Ala	GAG	Glu	GGG	Gly	G

6. TRANSLATION (PROTEIN SYNTHESIS)

- Translation is the process of synthesis of **polypeptide (protein)** using the information from mRNA.
- Occurs in **cytoplasm** on **ribosomes**.

TRANSLATION PROCESS (3D VIEW)

1 Initiation



2 Elongation



3 Termination



Amino Acid



tRNA



mRNA



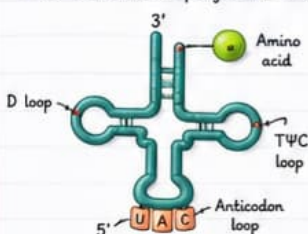
Ribosome

STEPS OF TRANSLATION

- Initiation**: Small ribosomal subunit binds to mRNA. Initiator tRNA (Met) binds to the start codon (AUG). Large subunit joins to form complete ribosome.
- Elongation**: tRNAs bring amino acids according to codons. Peptide bond forms between adjacent amino acids.
- Termination**: When ribosome reaches a stop codon (UAA / UAG / UGA), release factors bind and polypeptide is released.

tRNA - ADAPTOR MOLECULE

- Small RNA molecule (73-93 nucleotides).
- Has cloverleaf structure.
- Anticodon loop pairs with codon in mRNA.
- 3' end carries specific amino acid.



EXAMPLE

mRNA codon : AUG
tRNA anticodon : UAC
Amino acid : Methionine

IMPORTANT POINTS

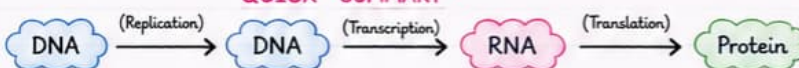
- ★ The flow of genetic information is DNA → RNA → Protein.
- ★ This flow is called **Central Dogma** of Molecular Biology.
- ★ Proteins determine the structure and function of an organism.

KEY EXAMPLES

- Hemoglobin → transports O_2 .
- Insulin → regulates blood sugar level.
- Enzymes → catalyze chemical reactions.



QUICK SUMMARY



"Understanding the code is understanding life."

Molecular Basis of Inheritance

7. RNA - THE MOLECULE OF DIVERSITY

- RNA (Ribonucleic Acid) is usually **single stranded**.
- It contains **Ribose sugar** and **Uracil (U)** in place of Thymine.
- It plays important roles in **protein synthesis**.

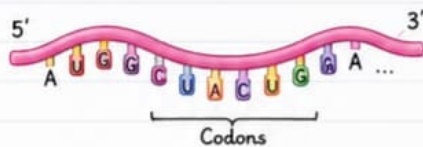
DIFFERENCE BETWEEN DNA AND RNA

Feature	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Strands	Double stranded	Single stranded
Location	Nucleus, Mitochondria, Chloroplasts	Nucleus, Cytoplasm (Ribosomes)
Function	Stores genetic information	Helps in protein synthesis

TYPES OF RNA

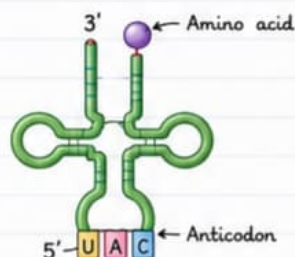
① mRNA (Messenger RNA)

- Carries genetic information from DNA to ribosome.
- It is linear.



② tRNA (Transfer RNA)

- Brings specific amino acids to ribosome.
- It has cloverleaf structure.



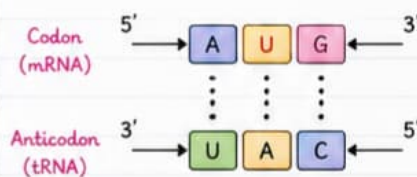
③ rRNA (Ribosomal RNA)

- Main structural and functional component of ribosomes.
- It helps in protein synthesis.



CODON - ANTICODON PAIRING

- Codon** → A group of three bases on mRNA that codes for an amino acid.
- Anticodon** → A group of three bases on tRNA that is complementary to the codon.



The pairing is complementary:



READING FRAME

- mRNA is read in triplets starting from AUG (start codon).
- The ribosome moves in one direction (5' → 3').



SPECIAL CODONS

- AUG** - Start codon (codes for Methionine)
- UAA** - Stop codon
- UAG** - Stop codon
- UGA** - Stop codon

These do not code for any amino acid.



SUMMARY MAP

MOLECULAR BASIS OF INHERITANCE

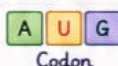
DNA

- Genetic material
- Double helix
- Made of nucleotides
- Bases: A, T, G, C
- Replicates (Semi-conservative)



GENETIC CODE

- Triplet code
- 64 codons (61 sense + 3 stop)
- Universal
- Non-overlapping
- Unambiguous



RNA

- mRNA - messenger
- tRNA - transfer
- rRNA - ribosomal
- Single stranded
- Contains Uracil (U)



PROTEIN SYNTHESIS

- Transcription (DNA → RNA)
- Translation (RNA → Protein)
- Occurs in ribosomes



KEY TAKEAWAY



The flow of genetic information is

DNA → RNA → PROTEIN.

This flow is the central dogma of molecular biology.

"Decode the code,
Understand life."
That's the beauty
of Biology."

IMPORTANT FACTS

- ✓ Nirenberg & Khorana deciphered the code.
- ✓ 61 codons code for amino acids.
- ✓ 3 codons are stop codons.
- ✓ Genetic code is nearly universal.

